6. Alternative methods of extracting copper

As the Earth's supply of high-grade copper ores continues to decline, alternative methods such as phytomining and bioleaching are becoming more commonly used to extract copper from low-grade ores.

Phytomining

Phytomining involves growing plants in soils that contain low-grade copper ores.

The plants absorb the copper ions through their roots and they are concentrated inside the plants' cells.

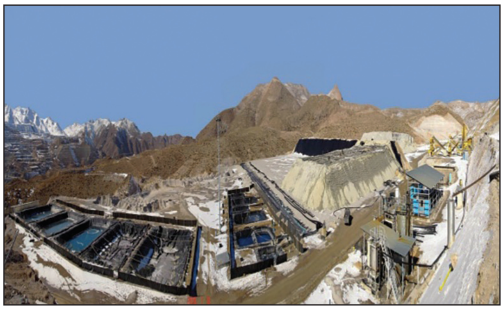
The plants are then harvested and burned. The ash that is left behind contains higher concentrations of copper ions than the low-grade ores in the soil.

The copper ions are washed out from the ash using sulfuric acid, forming a concentrated solution of copper(II) sulfate.

Scrap iron is then added to the copper(II) sulfate solution to allow the copper to be collected.

The main drawback to phytomining as an alternative method of copper extraction is that the process is dependent on the successful growth of the plants used for the extraction.

Bioleaching



Bioleaching is a very simple process that uses bacteria to extract copper from the minerals found within the lowest grade copper ores.

The bacteria are added to tanks of water containing the ores. There they feed on the nutrients found within the minerals in these ores, causing the copper ions to separate out.

An acidic solution is formed during the process. This contains a higher concentration of copper ions than the low-grade ores.

The solution formed is called a leachate, which is why the process is called bioleaching.

Copper is collected from the leachate solution by mixing it with iron.

The main drawback to bioleaching as an alternative method of copper extraction is that the process is extremely slow and therefore not efficient.

Examiner only

(a) (i) Tick (✓) the box next to the correct statement. [1]

phytomining uses lower grade ores than bioleaching	☐
bioleaching conserves supplies of the lowest grade metal ores	☐
phytomining conserves supplies of high-grade metal ores	☐
bioleaching uses high-grade metal ores	☐
phytomining uses the lowest grade metal ores	☐

(ii) State whether you agree with the following statement. Give a reason for your answer. [1]

'Phytomining and bioleaching both involve biological processes'

Agree **Yes** / **No**

Reason ....

....

(iii) Suggest a reason why phytomining might **not** be a suitable method for extracting metals in some countries, even if they have supplies of the low-grade metal ores. [1]

....

....

(iv) Although effective in extracting copper, suggest a reason why bioleaching is an extremely slow process. [1]

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Examiner only

(b) Explain why iron can be used in the last stage of both the phytomining and bioleaching methods of copper extraction. [2]

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(c) Low-grade copper ores commonly contain the mineral cuprite. Cuprous oxide is an old name for the oxide of copper found in cuprite.

A sample of cuprous oxide was found to contain 2.54 g of copper and 0.32 g of oxygen.

Use this information to find the simplest formula of cuprous oxide.

You **must** show your working. [3]

$A_r(\text{Cu}) = 63.5$ $\qquad$ $A_r(\text{O}) = 16$

Simplest formula ....

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